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import cv2
import numpy as np

def main():
    video_path = "t1.mp4"    # keep video in same folder

    cap = cv2.VideoCapture(video_path)

    if not cap.isOpened():
        print("Error: Cannot open video")
        return

    ret, first_frame = cap.read()
    if not ret:
        print("Error: Cannot read video")
        return

    prev_gray = cv2.cvtColor(first_frame, cv2.COLOR_BGR2GRAY)
    prev_gray = cv2.GaussianBlur(prev_gray, (15,15), 0)

    print("Press 'q' to quit")

    while True:
        ret, frame = cap.read()
        if not ret:
            break

        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
        gray = cv2.GaussianBlur(gray, (15,15), 0)

        diff = cv2.absdiff(prev_gray, gray)

        thresh = cv2.threshold(diff, 15, 255, cv2.THRESH_BINARY)[1]
        thresh = cv2.dilate(thresh, None, iterations=3)

        contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL,
cv2.CHAIN_APPROX_SIMPLE)

        for c in contours:
            if cv2.contourArea(c) > 200:
                x,y,w,h = cv2.boundingRect(c)
                cv2.rectangle(frame, (x,y), (x+w,y+h), (0,255,0), 2)

        cv2.imshow("Motion", frame)

        prev_gray = gray

        if cv2.waitKey(30) & 0xFF == ord('q'):
            break

```

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cap.release()  
cv2.destroyAllWindows()
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```
if __name__ == "__main__":  
    main()
```

Press 'q' to quit